

PURUSHOTTAM KUMAR

Cybersecurity Analyst

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PROFESSIONAL SUMMARY

Second-year B.Tech CSE (Cybersecurity) student with hands-on experience across offensive and defensive security tooling, network programming, and full-stack development. Shipped 7+ projects spanning a machine-learning-assisted intrusion detection system, an AI-powered code reviewer, and multiple Python network utilities — designed, coded, and validated end-to-end with automated tests.

EDUCATION

Central University of Jammu 2025 – 2029

B.Tech in Computer Science & Engineering (Cybersecurity) · SGPA: 8.73

TECHNICAL SKILLS

Languages: Python, C

Security: Network security fundamentals, OWASP Top 10 concepts, packet analysis (Scapy), signature & ML-based anomaly detection

Networking: Socket programming, TCP/IP, FTP, port scanning

Libraries / Tools: Scapy, Scikit-learn, socket, threading, struct, React

Platforms: Linux (Bash), Git/GitHub, Cloudflare Workers

CERTIFICATIONS & ACHIEVEMENTS

- Cisco — Networking Basics
- Cisco — Networking Devices & Basic Configuration
- Cisco — Introduction to Modern AI
- Snyk CTF — Ranked 328 / 1,500 Teams

PROJECTS

Network Intrusion Detection System (IDS) —

github.com/purushottamcuj62045-alt/IDS-Intrusion-detection-system

- Built a real-time IDS in Python using Scapy for packet capture and Scikit-learn (Isolation Forest) for ML anomaly detection, alongside signature-based rules for known attack patterns.
- Detects SYN floods and port scans with 1.00 confidence; logs structured JSON alerts (timestamp, threat type, source/destination IP, confidence). Validated with a 4-test automated suite — 4/4 passed.

Pluton — AI Code Reviewer — github.com/purushottamcuj62045-alt/Pluton

- Built a web app that auto-detects a code snippet's language and scores it on performance, security, readability, and maintainability, using React and TypeScript, deployed on Cloudflare Workers.

TCP Port Scanner

- Wrote a multi-threaded TCP port scanner in Python with service banner grabbing and timeout handling for efficient, low-noise reconnaissance.

FTP Server & Educational Password Auditing Tool

- Built an FTP server in Python implementing core file-transfer operations over sockets, and an educational tool to study password security and brute-force defense mechanisms.

AREAS OF INTEREST

Cybersecurity · Network Security · Ethical Hacking · Socket Programming · Systems Programming · Security Research